ON THE 7-RANK OF A PUTATIVE STRONGLY REGULAR GRAPH WITH PARAMETERS $(99, 14, 1, 2)$

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ABSTRACT. Let A be the adjacency matrix of a hypothetical strongly regular graph with parameters $(99, 14, 1, 2)$, the object of Conway's 99-graph problem. The only interesting prime for the p-ranks of A is 7, and the known bound is $\operatorname{rank}_7(A - 3I) \leq 45$. We prove that $\operatorname{rank}_7(A - 3I)$ is odd and that

$$23 \leq \operatorname{rank}_7(A - 3I) \leq 43.$$

The upper bound comes from the genus of the integral eigenvalue lattice $\mathbb{Z}^{99} \cap \ker(A + 4I)$, through the oddity formula. The lower bound comes from an exact local formula: the 7-rank is 15 plus the 7-rank of an explicit 84×84 matrix built from the second neighbourhood of any vertex. A small exact computation raises the lower bound to 25.

1. INTRODUCTION

A strongly regular graph Γ with parameters $(99, 14, 1, 2)$ has spectrum 14^1, 3^{54}, $(-4)^{44}$. Its adjacency matrix satisfies

$$A^2 + A - 12I = 2J, \qquad \text{that is} \qquad (A - 3I)(A + 4I) = 2J. \tag{1}$$

Whether such a graph exists is one of Conway's \$1000 problems [2]. The p-ranks of $A - \theta I$ are classical invariants of strongly regular graphs [1]. For primes p not dividing $r - s = 7$ these ranks are determined by the parameters (see [1]), so $p = 7$ is the only interesting prime here. Brouwer and van Eijl [1] prove $\operatorname{rank}_p(A - rI) \leq \min(f, g) + 1$, which gives $\operatorname{rank}_7(A - 3I) \leq 45$. We write

$$k_7 := \operatorname{rank}_7(A - 3I) - 15.$$

Theorem 1. *If Γ exists, then k_7 is even and $8 \leq k_7 \leq 28$, that is*

$$\operatorname{rank}_7(A - 3I) \in \{23, 25, \ldots, 43\}.$$

Moreover $k_7 = \operatorname{rank}_7(M_x)$ for every vertex x, where M_x is the 84×84 matrix of Proposition 5.

The upper bound and the parity are proved in Section 2. The local formula and the lower bound are proved in Sections 3 and 4. Remark 9 explains how a small exhaustive computation excludes $k_7 = 8$ as well.

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2. The genus of the eigenvalue lattice

Let $\Lambda = \mathbb{Z}^{99} \cap \ker(A + 4I)$. It is a primitive sublattice of $\mathbb{Z}^{99}$ of rank 44. The orthogonal projection onto $\ker(A + 4I)$ is

$$E = \frac{27I - 9A + J}{63}.$$

We use two standard facts.

(F1) For a primitive sublattice $L \subseteq \mathbb{Z}^n$, the orthogonal projection of $\mathbb{Z}^n$ onto $L \otimes \mathbb{Q}$ is the dual lattice $L^{\#}$.

(F2) A primitive sublattice L of a unimodular lattice and its orthogonal complement have isomorphic discriminant groups [4, Prop. 1.6.1].

Lemma 2. *Λ is even, and $\Lambda^{\#}/\Lambda \cong \mathbb{Z}/9 \oplus (\mathbb{Z}/7)^{\alpha}$ with $\alpha = \operatorname{rank}_7(A - 3I) - 1 = 14 + k_7$. In particular $\det \Lambda = 9 \cdot 7^{\alpha}$.*

Proof. Evenness. Every $u \in \Lambda$ is orthogonal to $\mathbf{1}$, so $u \cdot u \equiv \sum u_i = 0 \pmod 2$.

Exponent. Since $63E$ is integral, (F1) gives $63\Lambda^{\#} = 63E(\mathbb{Z}^{99}) \subseteq \Lambda$. So the discriminant group is killed by 63, and Λ is unimodular at every prime other than 3 and 7.

The 3-part. By (F2) it equals the 3-part of the discriminant group of $\Lambda' = \mathbb{Z}^{99} \cap (\mathbb{Q}\mathbf{1} \oplus \ker(A - 3I))$. Write $u \in \Lambda'$ as $u = c\mathbf{1} + w$ with $(A - 3I)w = 0$. Then $(A - 3I)u = 11c\mathbf{1}$ is integral, so $11c \in \mathbb{Z}$, and Λ' contains $\mathbb{Z}\mathbf{1} \oplus \Lambda_3$ with index dividing 11, where $\Lambda_3 = \mathbb{Z}^{99} \cap \ker(A - 3I)$. The projection onto $\ker(A - 3I)$ is $(44I + 11A - 2J)/77$. As above, $77\Lambda_3^{\#} \subseteq \Lambda_3$, so Λ_3 is unimodular at 3. Hence the 3-part of the discriminant group of Λ' is that of $\mathbb{Z}\mathbf{1}$, namely $\mathbb{Z}/9$.

The 7-part. Let $W = \mathbb{Z}^{99} \cap \mathbf{1}^{\perp}$. On W we have $J = 0$, hence $A - 3I = -7E$ by the formula for E. Since $[\mathbb{Z}^{99} : W \oplus \mathbb{Z}\mathbf{1}] = 99$ is prime to 7, we get $\mathbb{Z}_7^{99} = W_7 \oplus \mathbb{Z}_7\mathbf{1}$ (subscripts denote $\otimes\mathbb{Z}_7$), and $(A - 3I)(W_7) = 7E(\mathbb{Z}_7^{99}) = 7\Lambda_7^{\#}$. Reducing modulo 7, the image of $A - 3I$ is spanned by $7\Lambda_7^{\#}$, which is $\Lambda_7^{\#}/\Lambda_7$ in $\mathbb{F}_7^{99}$, together with $(A - 3I)\mathbf{1} = 11 \cdot \mathbf{1}$. Moreover $\mathbf{1}$ is not in the reduction of $7\Lambda_7^{\#}$, because $\mathbf{1} \cdot \mathbf{1} = 99 \not\equiv 0$ while $7\Lambda_7^{\#} \perp \mathbf{1}$. Hence $\operatorname{rank}_7(A - 3I) = \dim_{\mathbb{F}_7}(\Lambda^{\#}/\Lambda)_7 + 1$. The 7-part has exponent 7, so it is $(\mathbb{Z}/7)^{\alpha}$ with $\alpha = \operatorname{rank}_7(A - 3I) - 1$. □

Theorem 3. *α is even and $\alpha \leq 42$; equivalently k_7 is even and $k_7 \leq 28$. More precisely, write $\Lambda \otimes \mathbb{Z}_7 = U \perp 7V$ with U, V unimodular over $\mathbb{Z}_7$, of ranks $30 - k_7$ and $14 + k_7$. Then $U \neq 0$, and $U \otimes \mathbb{F}_7$ is not a hyperbolic quadratic space.*

Proof. We use the oddity formula of Conway and Sloane [3, Ch. 15, §7]. For a lattice of signature σ,

$$\sigma + \sum_{p \text{ odd}} p\text{-excess} \equiv \text{oddity} \pmod 8.$$

For an odd prime p, a Jordan constituent of scale $q = p^j$, dimension n and sign ε contributes $n(q-1)+4[\,q$ is not a square and $\varepsilon = -1\,]$ to the p-excess.

Here $\sigma = 44$. Since Λ is even of odd determinant, it is unimodular and even at 2, and its oddity is 0. By Lemma 2 the 3-adic constituents are a unimodular one and $9^{\pm 1}$; the latter contributes $1\cdot(9-1) = 8 \equiv 0$ whatever its sign, as 9 is a square. The 7-adic constituents are U and $7V$, with $\dim V = \alpha$; the constituent $7V$ contributes $6\alpha + 4\delta$, where $\delta = 1$ if $\left(\frac{\det V}{7}\right) = -1$ and $\delta = 0$ otherwise. The oddity formula therefore reads

$$44 + 6\alpha + 4\delta \equiv 0 \pmod 8.$$

Hence α is even; write $\alpha = 14+2m$, so that $k_7 = 2m$. Then $44+84+12m+4\delta \equiv 4(m+\delta) \pmod 8$, and so $\delta \equiv m \pmod 2$.

Since $\det \Lambda = 9 \cdot 7^\alpha$, we have $\left(\frac{\det U}{7}\right)\left(\frac{\det V}{7}\right) = \left(\frac{9}{7}\right) = 1$.

Suppose first that $U = 0$, that is $k_7 = 30$ and $m = 15$. Then $\left(\frac{\det V}{7}\right) = 1$, so $\delta = 0$, contradicting $\delta \equiv m = 15$. Thus $U \neq 0$ and $k_7 \leq 28$.

In general, $\left(\frac{\det U}{7}\right) = \left(\frac{\det V}{7}\right) = (-1)^m$. A quadratic space of even dimension $2t = 30 - k_7$ over $\mathbb{F}_7$ is hyperbolic if and only if its determinant is $(-1)^t$ modulo squares. Here $t = 15 - m$ and $\left(\frac{-1}{7}\right) = -1$, so hyperbolicity would require $(-1)^m = (-1)^{15-m}$, which never holds. □

Remark 4. The same computation applied to graphs that exist is consistent. For the Petersen graph ($r - s = 3$), the lattice $\mathbb{Z}^{10} \cap \ker(A + 2I)$ is entirely 3-divisible in the sense above ($U = 0$). This is allowed there, because the unit part of its determinant, 5, is a non-square modulo 3. For $(99, 14, 1, 2)$ the unit part 9 is a square modulo 7, and this is what forbids $U = 0$.

3. An exact local formula

Fix a vertex x. Its neighbourhood consists of 7 disjoint edges, the *mate pairs*. We write $\bar{a}$ for the mate of $a \in N(x)$. Since $\mu = 2$, every vertex at distance 2 from x is adjacent to exactly two vertices of $N(x)$. These two vertices are not mates, by $\lambda = 1$, and every non-mate pair occurs exactly once. So the 84 vertices at distance 2 are labelled by the 84 non-mate pairs $\{a, b\} \subset N(x)$.

Let H be the adjacency matrix of the second neighbourhood, indexed by these pairs. For two pairs y, z let $s(y, z) = |y \cap z|$, and let $p(y, z)$ be the number of $(i, j) \in y \times z$ with $j = \bar{i}$.

Proposition 5. *Let $M_x = H+4I+3S+P'+5J$ over $\mathbb{F}_7$, where $S = (s(y, z))$ and $P' = (p(y, z))$. Then*

$$\operatorname{rank}_7(A - 3I) = 15 + \operatorname{rank}_7(M_x).$$

In particular $\operatorname{rank}_7(M_x) = k_7$ *does not depend on x.*

Proof. Order the vertices as $\{x\} \cup N(x)$ followed by the second neighbourhood. Then $A - 3I = \begin{pmatrix} F - 3I & B \\ B^T & H - 3I \end{pmatrix}$, where F is the adjacency matrix

of the closed neighbourhood (a windmill), and the column of B at $y = \{a, b\}$ is $e_a + e_b$. A direct computation gives $\det(F - 3I) = 2^{23}$, so $F - 3I$ is invertible over $\mathbb{F}_7$. Moreover the block of $(F - 3I)^{-1}$ on $N(x)$ is

$$G = -\tfrac{3}{8}I - \tfrac{1}{8}P + \tfrac{1}{16}J,$$

where P is the mate involution. To check this, note that the rows of $F - 3I$ on $N(x)$ are $(1,\ P - 3I)$, and multiply out using $P^2 = I$, $PJ = J$.

So $\operatorname{rank}_7(A - 3I) = 15 + \operatorname{rank}_7(H - 3I - B^T GB)$. Now $(B^T GB)_{yz} = \sum_{i \in y,\, j \in z} G_{ij}$. Modulo 7 we have $-\frac{3}{8} \equiv 4$, $-\frac{1}{8} \equiv 6$ and $\frac{1}{16} \equiv 4$. Since $|y| = |z| = 2$, this gives $(B^T GB)_{yz} \equiv 4s + 6p + 4 \cdot 4 = 4s + 6p + 2$. Hence

$$H - 3I - B^T GB \equiv H - 3I + 3S + P' + 5J \equiv M_x \pmod 7. \qquad \square$$

4. Lower bounds from stars

For $a \in N(x)$, the *star* of a is the set of the 12 pairs $\{a, b\}$ with $b \notin \{a, \bar{a}\}$. Two pairs $\{a, b\}$ and $\{a, c\}$ are adjacent if and only if they are matched by the perfect matching τ_a on the 12 elements b defined by the triangles through a (the edge $a\{a, b\}$ lies in a unique triangle). The six mate pairs $\{b, \bar{b}\}$ ($b \neq a, \bar{a}$) are called *classes*.

The *class graph* of τ_a is the 2-regular multigraph on the six classes with one edge $\{\text{class}(b), \text{class}(\tau_a b)\}$ for each pair of τ_a. A loop arises when $\tau_a(b) = \bar{b}$. Let $e(\tau_a)$ be the number of cycles of even length in the class graph; a double edge is a cycle of length 2, and loops are not counted. Clearly $e(\tau_a) \leq 3$.

Lemma 6. *The principal submatrix of M_x on the star of a has 7-rank $11 - 2e(\tau_a)$.*

Proof. On the star, $S = I + J$, and $H + P' =: N$ is the adjacency matrix of the union of the two perfect matchings τ_a and $b \mapsto \bar{b}$. So the block is $N + 7I + 8J \equiv N + J$.

The union of two perfect matchings is a disjoint union of alternating cycles. A loop of the class graph gives a double edge $\left(\begin{smallmatrix} 0 & 2 \\ 2 & 0 \end{smallmatrix}\right)$, which is invertible modulo 7. A cycle of length $m \geq 2$ of the class graph gives one alternating cycle of length $2m$. Over $\mathbb{F}_7$, a vector in the kernel of the adjacency matrix of the cycle C_{2m} satisfies $v_{i+2} = -v_i$. Such vectors exist (forming a space of dimension 2) exactly when $4 \mid 2m$, that is when m is even, and then they have sum 0. Hence $\dim \ker_7 N = 2e(\tau_a)$.

Finally $N\mathbf{1} = 2 \cdot \mathbf{1}$, so $(N + J)\mathbf{1} = 14 \cdot \mathbf{1} \equiv 0$. If $(N + J)v = 0$, then $w = v + \frac{1}{2}(\sum v)\mathbf{1}$ lies in $\ker N$. Hence $\ker_7(N + J) = \ker_7 N \oplus \mathbb{F}_7\mathbf{1}$ has dimension $2e(\tau_a) + 1$, and the rank is $12 - 2e(\tau_a) - 1$. $\square$

Corollary 7. *$k_7 \geq 11 - 2e(\tau_a)$ for every vertex x and every $a \in N(x)$. In particular $k_7 \geq 6$.*

Proof. A principal submatrix has rank at most $\operatorname{rank}_7(M_x) = k_7$. Then use $e \leq 3$ and the parity of k_7 (Theorem 3). $\square$

Proposition 8. *$k_7 \neq 6$. Hence $k_7 \geq 8$ and $\operatorname{rank}_7(A - 3I) \geq 23$.*

Proof. Suppose $k_7 = 6$. By Corollary 7 every star block has rank at most 6, hence rank exactly 5, and $e(\tau_a) = 3$: the class graph of every τ_a consists of three double edges.

Since M_x is symmetric of rank 6 over $\mathbb{F}_7$, we can write $M_x = V^T Q V$ with V of size 6×84 and Q invertible. Let V_a be the columns of V indexed by the star of a, and U_a their span. Since $\operatorname{rank}(V_a^T Q V_a) = 5$, we have $\dim U_a \in \{5, 6\}$. If $\dim U_a = 6$, then $V_a^T Q V_a$ would have rank 6. Hence $\dim U_a = 5$ and $Q|_{U_a}$ is nondegenerate. Consequently every vector c in the kernel of the star block satisfies $V_a c = 0$, that is,

$$\sum_y c_y \, (M_x)_{y,\cdot} = 0 \quad \text{on all 84 columns.}$$

Take a double edge of the class graph: classes $\{\ell_1, \ell_4\}$ and $\{\ell_2, \ell_3\}$ with $\tau_a(\ell_1) = \ell_2$, $\tau_a(\ell_3) = \ell_4$, $\bar{\ell}_2 = \ell_3$ and $\bar{\ell}_4 = \ell_1$. In the proof of Lemma 6, the columns ℓ_1 and ℓ_3 of N are both $e_{\ell_2} + e_{\ell_4}$. So $c = e_{\ell_1} - e_{\ell_3}$ is in the kernel of the star block, and the rows of M_x at $y = \{a, \ell_1\}$ and $y' = \{a, \ell_3\}$ coincide.

Choose $q \in N(x) \setminus \{a, \bar{a}, \ell_1, \bar{\ell}_1, \ell_3\}$ (nine choices) and let $z = \{\ell_1, q\}$. By the definition of M_x,

$$(M_x)_{yz} = H_{yz} + 3 + 5, \qquad (M_x)_{y'z} = H_{y'z} + [\, q = \ell_2 \,] + 5.$$

Here we used $s(y, z) = 1$, $p(y, z) = 0$ (as $q \neq \bar{a}$ and $\bar{\ell}_1 \notin z$), $s(y', z) = 0$, and $p(y', z) = [q = \bar{\ell}_3] = [q = \ell_2]$. Equality modulo 7 gives $H_{yz} - H_{y'z} \equiv [\, q = \ell_2 \,] - 3 \in \{4, 5\} \pmod 7$, which is impossible since $H_{yz}, H_{y'z} \in \{0, 1\}$. □

Remark 9 ($k_7 \neq 8$). The argument of Proposition 8 extends. If $k_7 = K$, the kernel vectors of a star block that are relations between complete rows of M_x form a space R_a of dimension at least $12 - K$. Each column z outside the star imposes a linear condition on R_a, and this condition depends only on the two (unknown) neighbours of z in the star, so it ranges over a small finite set. For $K = 8$ an exhaustive computation over $\mathbb{F}_7$ shows:

- for two of the three admissible star types, no subspace of the required dimension meets all the constraint sets;
- for the third type, exactly 32 subspaces remain. Adding the degree and labelling constraints of the second neighbourhood, each of the 32 resulting Boolean instances is refuted by unit propagation alone.

Hence $k_7 \geq 10$, that is $\operatorname{rank}_7(A - 3I) \geq 25$. We regard this as computer-assisted, although the computation is tiny. For $K = 10$ the one-star method no longer excludes all types.

Remark 10. A proof that $\operatorname{rank}_7(A - 3I) \leq 21$, that is $k_7 \leq 6$, would therefore prove that no srg(99, 14, 1, 2) exists.

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